

EXAM PRACTICE QUESTIONS

- 1 Consider the following code that processes an input string:

Java

```
public static void content(String text) {  
    String value = "";  
    int pos = 0;  
    while (pos < text.length()) {  
        String letter = text.substring(pos, pos + 1);  
        if (letter.equals(",")) {  
            System.out.println(value);  
            value = "";  
        } else {  
            value += letter;  
        }  
        pos++;  
    }  
    if (!value.isEmpty()) {  
        System.out.println(value);  
    }  
}
```

Python

```
def content(text):  
    value = ""  
    pos = 0  
    while pos < len(text):  
        letter = text[pos:pos+1]  
        if letter == ",":  
            print(value)  
            value = ""  
        else:  
            value = value + letter  
        pos = pos + 1  
    if value:  
        print(value)
```

- a Trace the algorithm for the input value of: "car,boat,ball" by copying and completing the following table, up to the variable pos becoming 7. [4]

VALUE	POS	LETTER	LETTER = ","	OUTPUT

- b Deduce the purpose of the algorithm. [3]
c The output is dependent on the exact format of the input. Identify two strings that would not generate the desired output. [2]

- 2 A group of friends play a round-robin tournament game of table tennis, where each person plays against each other player for 5 minutes at a time. The information from the tournament is recorded in parallel arrays (lists), `NAMES` and `SCORES`, examples of which are shown below.

When two people have their round against each other, each player records the points they win in their assigned row, in the column for the person they are playing against. For example, in the data set below, when Annabelle played against Jack, `SCORES[0][3]` shows that Annabelle scored 4 points, and `SCORES[3][0]` shows that Jack scored 8 points.

		SCORES					
		NAMES					
		0	1	2	3	4	5
0	Annabelle	0	3	4	4	3	5
1	Benjamin	5	0	3	5	3	3
2	Claire	7	5	0	4	3	5
3	Jack	8	6	4	0	5	4
4	Fran	2	1	4	5	0	2
5	Mark	2	3	2	5	4	0

- Construct an algorithm that will declare and populate the `NAMES` array (list), based on the example data above. [2]
 - Identify the data type of the elements in the `SCORES` array (list). [1]
 - Identify the winning player in the game between Claire and Benjamin. [1]
 - Construct an algorithm that will print the names of players for every round, each player's respective scores and who the winner was. If the game was a tie, output an appropriate message. For example:
Annabelle scored 3 vs Benjamin scored 5: Benjamin won
Annabelle scored 4 vs Claire scored 7: Claire won
... [6]
 - Construct an algorithm that will declare two new parallel arrays (lists) to keep track of a leaderboard based on net points each player has won. Perform the calculations necessary to populate these arrays. The two parallel arrays should be `LeaderName` and `LeaderPoints`. Each player's element in the `LeaderPoints` array should be the total of points they won minus the total of points they conceded.
For example, the first two rows using the data above would be:

LeaderName	LeaderPoints	
Annabelle	-5	(Annabelle won 19 points but conceded 24)
Benjamin	1	(Benjamin won 19 points and conceded 18)

[8]
 - Construct an algorithm that will sort the parallel arrays (lists) of `LeaderName` and `LeaderPoints` in descending order of `LeaderPoints`. [6]
 - Construct an algorithm that will save the parallel arrays (lists) of `NAMES` and `SCORES` to `names.txt` and `scores.txt` text files. [4]
- 3 A company has exported its sales data from a spreadsheet to a couple of text files. `NAMES.txt` contains a list of the names of its salespeople and `SALES.txt` contains a list of the total sales made by each person in the last month. For example, the first few lines of each file may resemble the following:
- | <code>NAMES.txt</code> | <code>SALES.txt</code> |
|------------------------|------------------------|
| Amina | 23424 |
| Carlos | 42549 |
| Emily | 52488 |
| Hao | 37562 |
| Isabella | 44770 |
- Construct an algorithm that will declare two parallel arrays or lists, `NAMES` and `SALES`, open the files and load their content into the respective array or list. You may assume there is a maximum of 1000 entries in the file. [4]
 - Construct an algorithm that will use a recursive quicksort to sort the two parallel arrays or lists by `SALES` in descending order. [8]
 - The content of the `NAMES` array (list) was iterated over with a loop, and its elements added into an empty stack (element 0 was the first to be added to the stack). What can be said about the order of the `NAMES` that will be popped off the stack in respect to their sales? [3]

- 4 a Define the term "queue". [2]
b Describe an application that uses a queue in a computer system. [2]
- 5 State the efficiency of the quicksort algorithm in Big O notation. [1]
- 6 Outline two uses of a stack in a computer system. [4]
- 7 Outline the differences between storing a queue in a linear form as opposed to a circular form. [4]
- 8 State which data type is best for storing a telephone number (e.g. 00352 661 008 990) and give two reasons. [3]
- 9 Arrays or linked lists can be used to implement stacks and queues.
a Describe the advantage of using an array to implement a stack or a queue. [1]
b Describe the advantage of using a linked list to implement a stack or a queue. [1]
- 10 The following list of numbers needs to be put into ascending order:
3, 11, 7, 2, 4, 1, 6
State the list that would be obtained after two iterations of a selection sort. [1]
- 11 Define the term "recursion". [1]
- 12 Construct a program to calculate the sum of numbers from 1 to n using recursion. [4]
- 13 State one advantage and one disadvantage of recursion vs iteration. [2]
- 14 Consider the code shown below.

Java

```
public static void manipulate(int n, int[] a) {
    a[0] = 15;
}
public static void main(String[] args) {
    int[] a = {0, 1};
    manipulate(3, a);
    for(int i = 0; i < a.length; i++) {
        System.out.println(a[i]);
    }
}
```

Python

```
def manipulate(n, a):
    a[0] = 15
a = [0, 1]
manipulate(3, a)
print(a)
```

- a State the scope of the variable n. [1]
- b An integer array (list) called b is initialized with the values {11,12,13,14}. The procedure manipulate is called again with the parameters 14 and b. Explain why the assignment a[0]=15 in the body of the procedure changes the values stored in the array (list) b. [3]
- 15 Consider the following data structure:

Java

```
int a = {1, 2, 3, 4, 5};
```

Python

```
a = [1, 2, 3, 4, 5]
```

- a State the name of the data structure. [1]
- b Outline how the data value 3 can be directly accessed. [1]

